

WHITE LESIONS OF ORAL CAVITY



LEARNING OUTCOMES

- **By the end of this chapter students will be able to:**
 1. Classify the white lesions of oral cavity.
 2. Describe the etiopathogenesis, complications and management of common White lesions affecting mouth.
 3. Explain the clinicopathologic features of leukoplakia, submucous fibrosis, candidiasis and lichen planus.
 4. Correlate the clinical and microscopic features of common white lesions to arrive at diagnosis.

CLASSIFICATION

1. Hereditary Conditions

- ▶ Leukoedema
- ▶ Intraepithelial Dyskeratosis
- ▶ White sponge nevus
- ▶ Follicular Keratosis

2. Reactive Lesions

- ▶ Frictional Hyperkeratosis
- ▶ Nicotinic stomatitis
- ▶ Smokeless tobacco lesions
- ▶ Hairy tongue

3. Pre-neoplastic Lesions

- ▶ **Leukoplakia (Epithelial Dysplasia)**
- ▶ Actinic Chelitis
- ▶ Actinic Keratosis

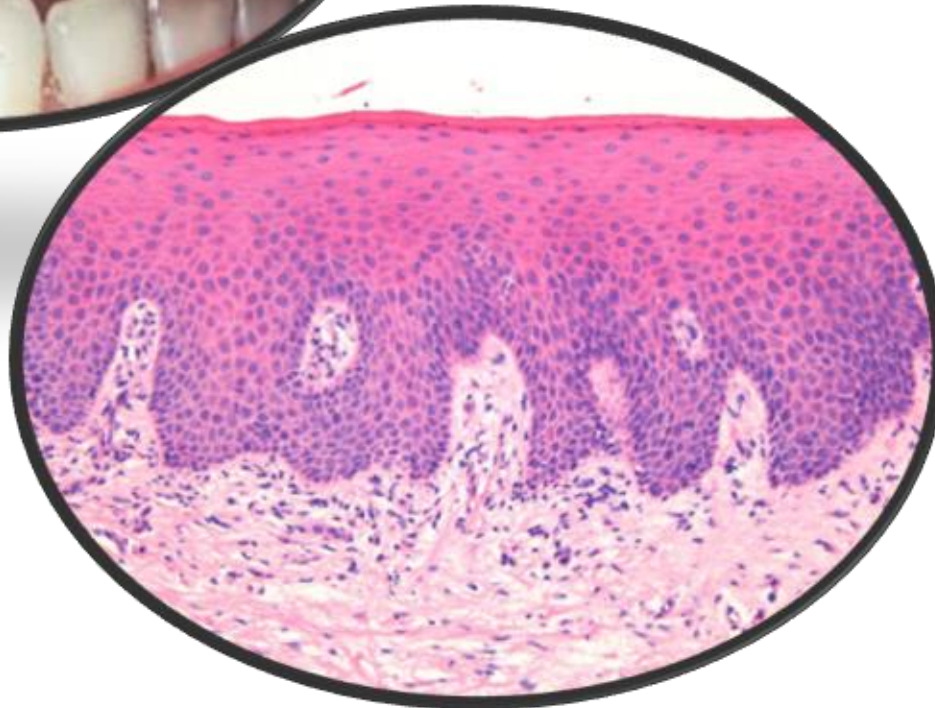
4. Non-epithelial yellow-white lesions

- ▶ **Oral Submucous Fibrosis**
- ▶ **Candidiasis**
- ▶ Fordyce's granules
- ▶ Ectopic lymphoid tissue
- ▶ Mucosal Burns

5. Other white lesions (Miscellaneous)

- ▶ **Lichen Planus**
- ▶ Lupus erythematosus
- ▶ Geographic tongue

CAUSES FOR DEVELOPMENT OF WHITE LESION



- ▶ Increased thickness of epithelium
- ▶ Abnormal Keratin production
- ▶ Restriction in blood supply
- ▶ Abnormal permeability of mucosa

POTENTIALLY MALIGNANT DISORDERS

PRECANCEROUS LESION

A morphologically altered tissue in which oral cancer is more likely to occur than in its apparently normal counterpart

- ✓ Leukoplakia
- ✓ Erythroplakia
- ✓ Mucosal changes associated with smoking habits
- ✓ Carcinoma in situ
- ✓ Bowen's disease
- ✓ Actinic keratosis, cheilitis and elastosis

PRECANCEROUS CONDITION

A generalized state associated with a significantly increased risk of cancer

- ✓ Oral submucous fibrosis
- ✓ Syphilis
- ✓ Sideropenic dysplasia
- ✓ Oral lichen planus
- ✓ Dyskeratosis congenita
- ✓ Lupus erythematosus

LEUKOEDEMA



- ▶ Filmy opalescence of the mucosa in early stages to grayish white cast with with a coarsely wrinkled surface.
- ▶ Bilateral lesion in buccal mucosa that disappears on stretching mucosa.
- ▶ Commonly noted among Negro men and women.
- ▶ No definite cause identified.
- ▶ No treatment necessary, it is a variant of normal mucosa.

FRICTIONAL HYPERKERATOSIS

Excessive keratin formation due to friction from rubbing of two surfaces against each other

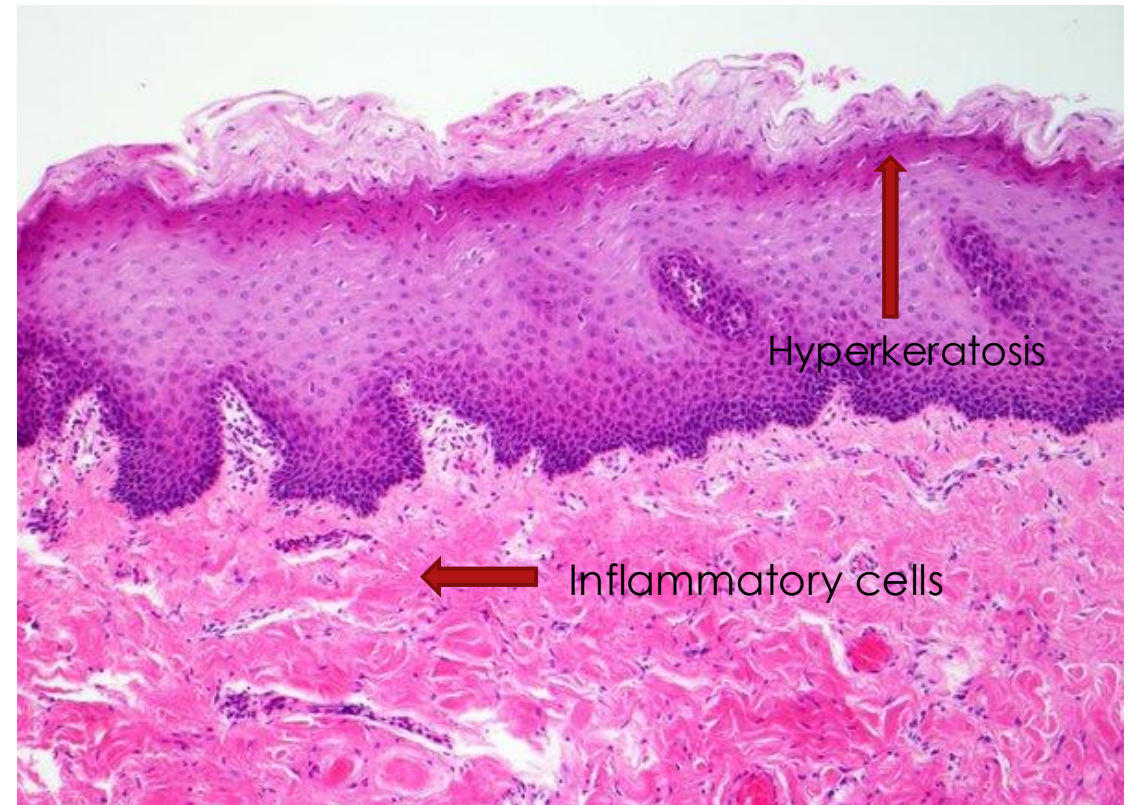


Histopathology

- ▶ Hyperkeratosis and few juxta-epithelial inflammatory cell

Treatment and Prognosis

- Removal of traumatic cause
- Discontinuation of habits
- Follow-up





LEUKOPLAKIA

Definition: A predominantly white lesion of oral cavity that cannot be characterized as any other definable lesion; some oral leukoplakia transform into cancer.



Premalignant Lesion / Potentially
Malignant lesion

Etiopathogeneisis of Leukoplakia

- Adaptive response of the oral mucosa to noxious stimuli.
- Contributory causes are tobacco, alcohol, candida infection, Human Papilloma virus, nutritional deficiency, trauma?.

Clinical features

- Affects middle aged and older population.
- Common in males.
- Common in vestibular area and buccal mucosa and tongue.**

Clinical Types of Leukoplakia

1. Homogenous: Smooth, Furrowed, Ulcerated
2. Non-homogenous: Nodulo-speckled



Speckled Leukoplakia



Smooth surface



Fissured/Furrowed surface



Ulcerated

Diagnosis of Leukoplakia

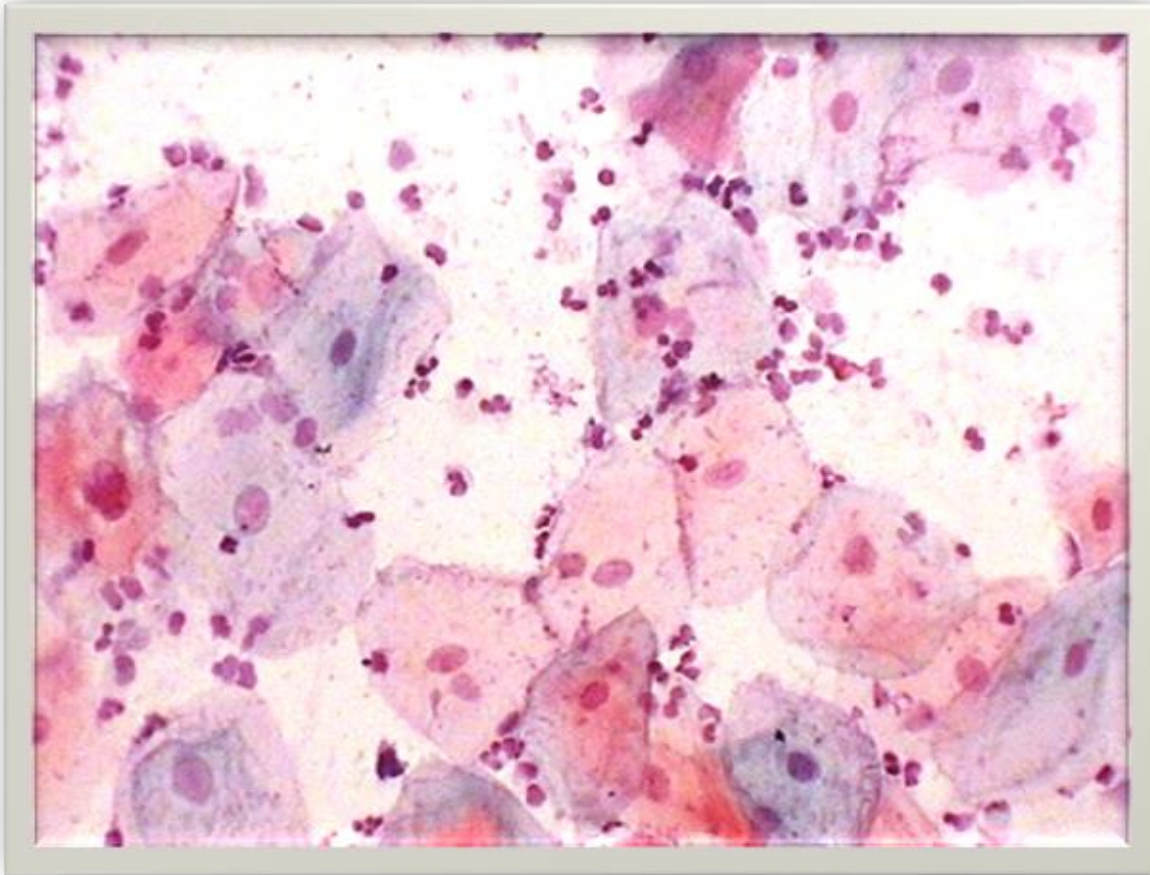
>>>>Cytology<<<<

>>>>Biopsy<<<<

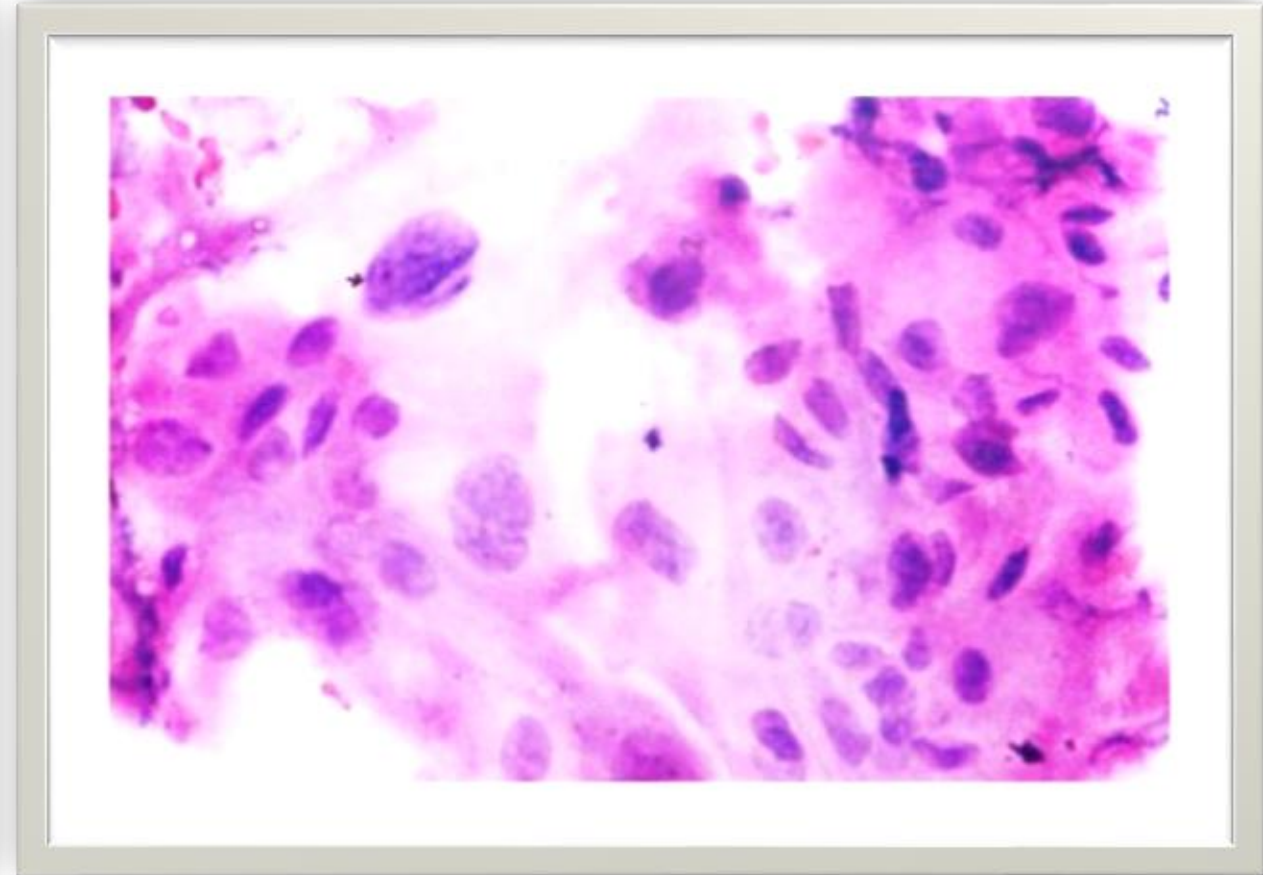


Cytology

Normal

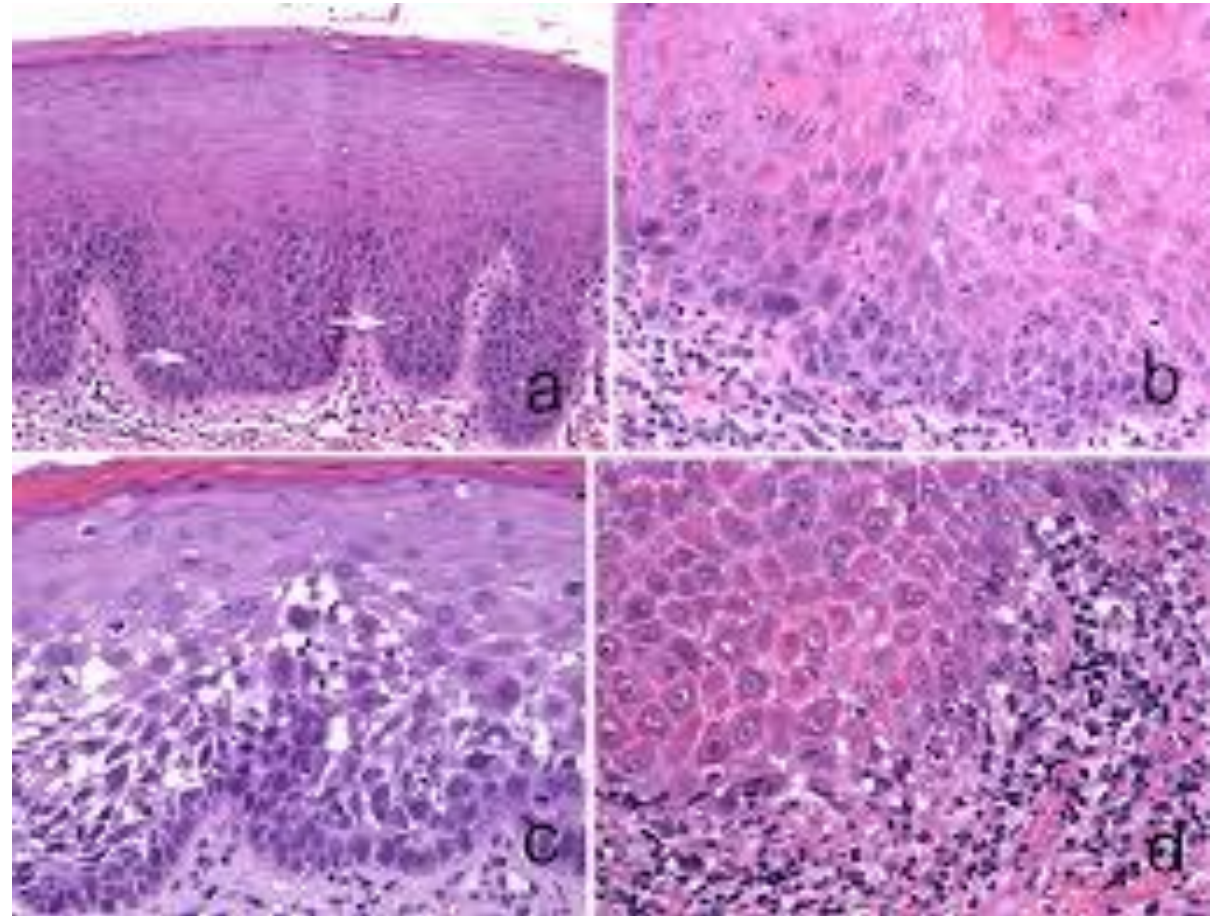


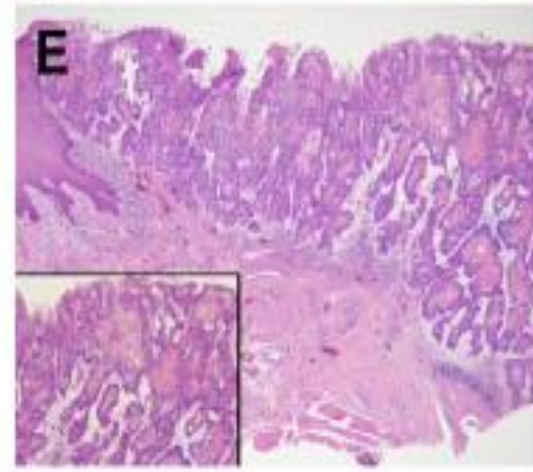
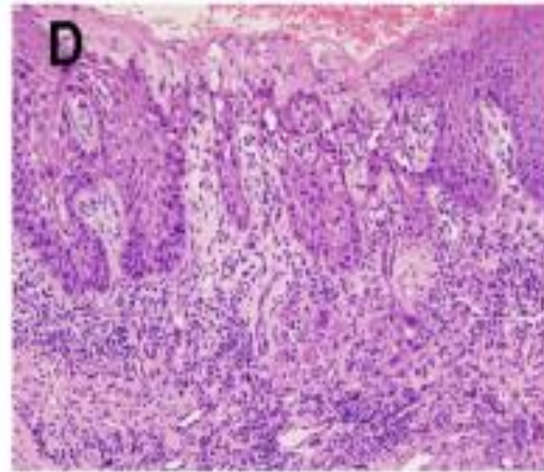
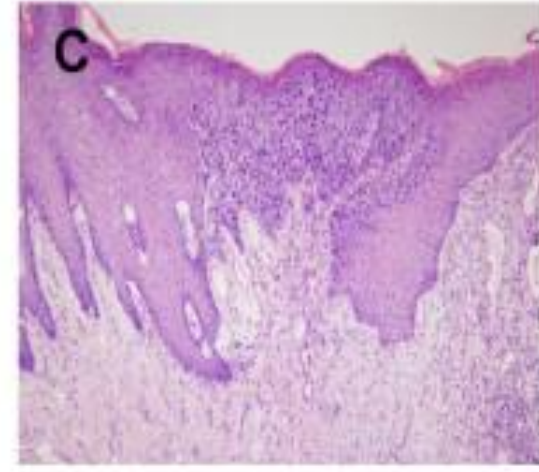
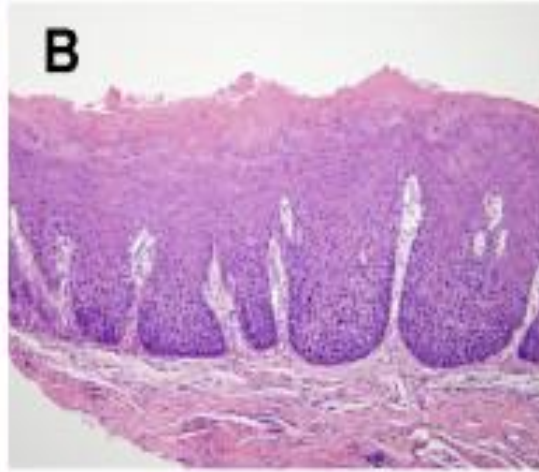
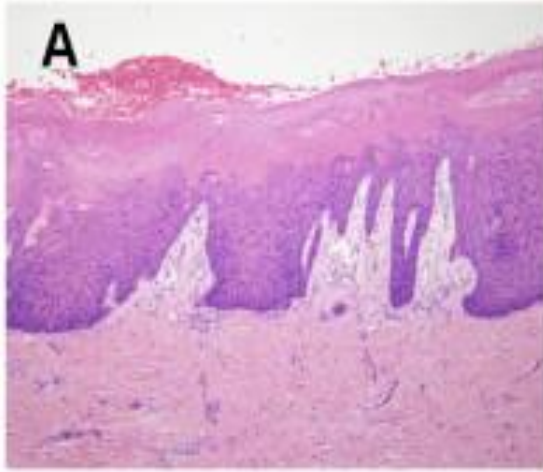
Dysplasia



Histopathology

1. Drop shaped epithelial rete ridges
2. Loss of stratification
3. Basal cell hyperplasia
4. Loss of cohesion
5. Increased mitosis
6. Abnormal mitotic figures
7. Individual cell keratinization
8. Pleomorphism of cells and nuclei
9. Hyperchromatism of nuclei
10. Increased nuclear-cytoplasmic ratio





Dysplasia of epithelium can be mild, moderate or severe (A, B & C).

Dysplasia involving entire thickness of epithelium, is called as Carcinoma in situ (D).

Treatment and Prognosis of Leukoplakia

Treatment

- ▶ Elimination of Cause
- ▶ Retinoids
- ▶ Surgical excision: scalpel excision, cryosurgery, electro-surgery, laser surgery.

Prognosis

- ▶ Untreated leucoplakia can progress to verrucous hyperplasia, verrucous carcinoma, micro-invasive carcinoma and squamous cell carcinoma.
- ▶ Leukoplakia in Floor of mouth & lateral border of tongue shows highest malignant transformation.



ORAL SUBMUCUS FIBROSIS (OSMF)



Premalignant Condition /
Potentially Malignant Condition

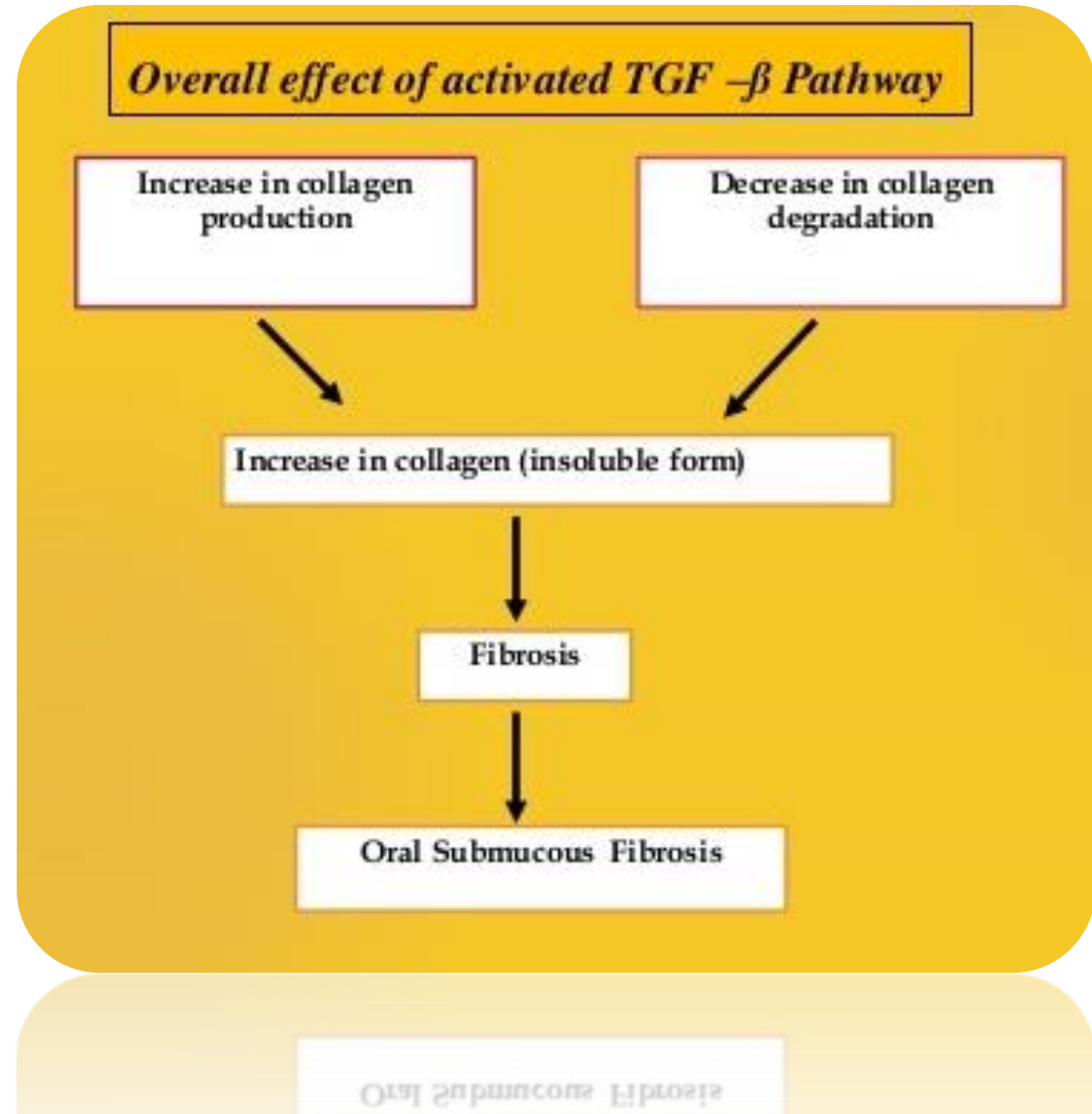


Etiology: Areca nut or Betel nut chewing

Pathogenesis

Contributing factors:

- Iron/folic acid deficiency,
- Pepper,
- Tobacco,
- Hypersensitivity



Clinical Features

Classification based on clinical features of OSMF

Haider et al. (2000)

Clinical staging:

- Stage 1: faucial bands only
- Stage 2: faucial & buccal bands
- Stage 3: faucial, buccal & labial bands

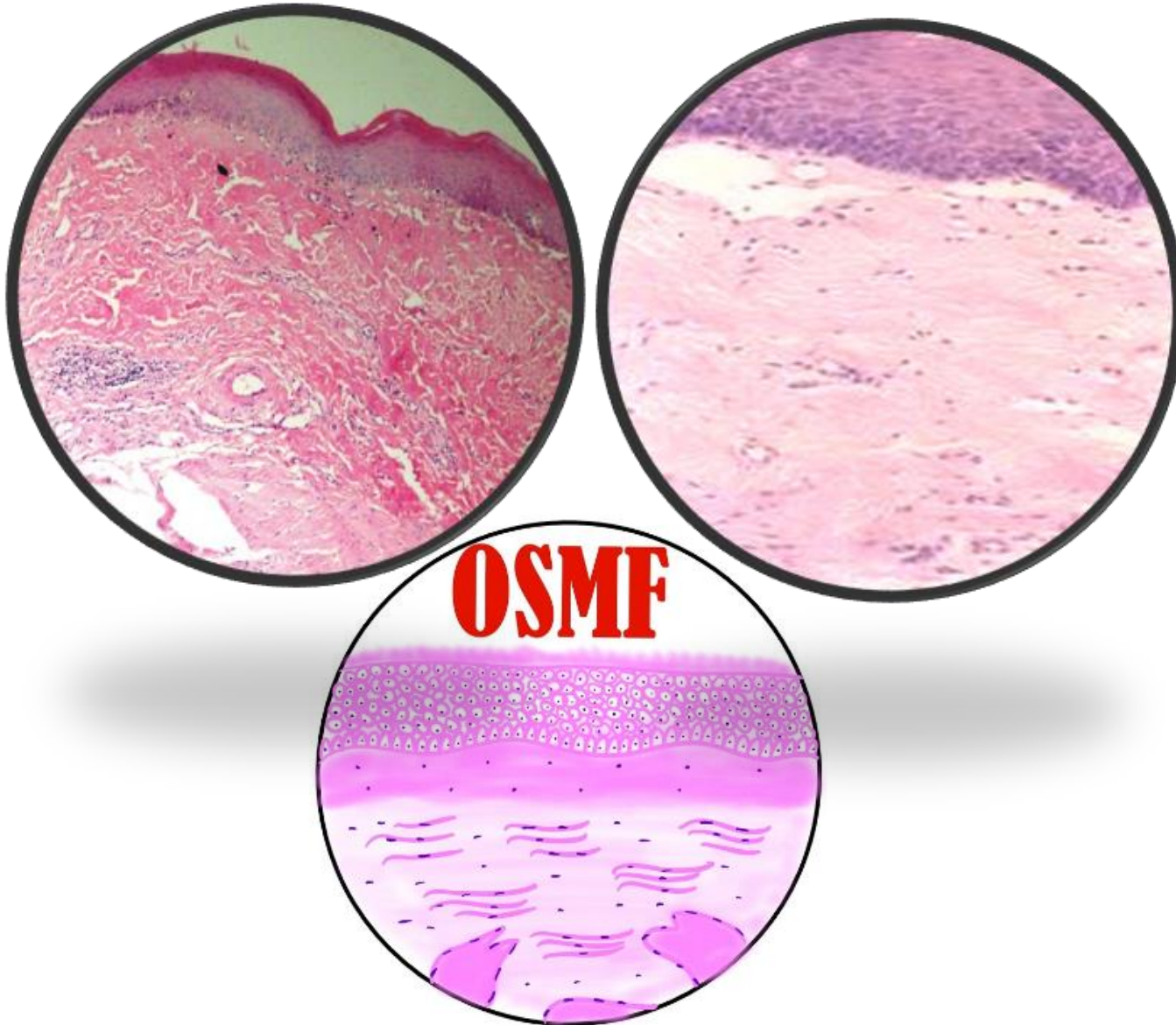
Functional staging:

- Stage 1: mouth opening >20 mm
- Stage 2: mouth opening 11-19 mm
- Stage 3: mouth opening <10 mm

Haider SM, Merchant AT, Fikree FF, Rahbar MH. Clinical and functional staging of oral submucous fibrosis. Br J Oral Maxillofac Surg. 2000 Feb;38(1):12-5.

- ▶ Common: 20-40 years.
- ▶ Typically seen in the oral cavity: extend into pharynx and oesophagus.
- ▶ Present as blanched mucosa.
- ▶ Oral mucosa loses its resilience and elasticity with time.
- ▶ Stiff mucosa due to fibrous bands.
- ▶ Fibrosis leads to trismus resulting in difficulty in eating.

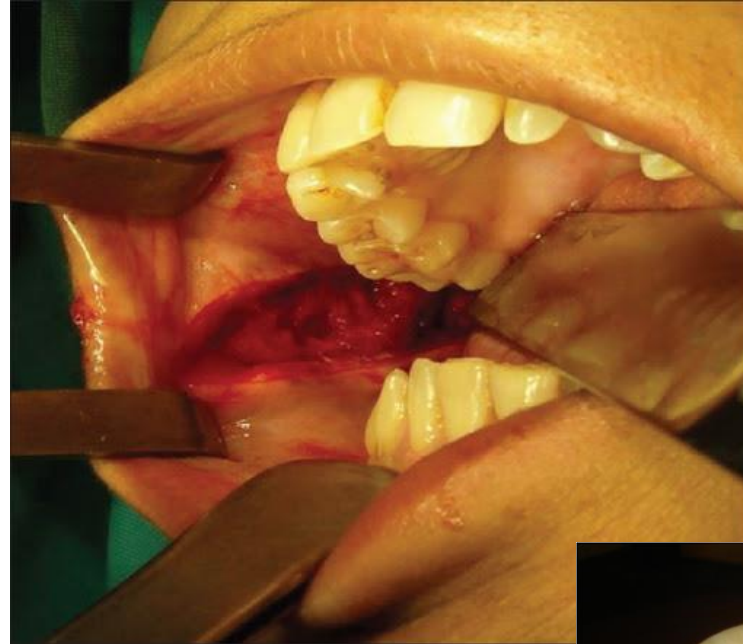
Histopathology of OSMF



- Sub-epithelial inflammatory infiltrate
- Fibrosis in the connective tissue and musculature
- Decreased vascularity due to fibrosis
- Epithelial dysplasia
- Atrophy of epithelium
- Hyalinization of connective tissue: in longer duration

Treatment and Prognosis

- ▶ Eliminate causative agent.
- ▶ Local injection of hyaluronidase, Chymotrypsin and dexamethasone.
- ▶ Surgical excision of fibrous bands.
- ▶ It is a premalignant condition, and if untreated can progress to squamous cell carcinoma.



CANDIDIASIS

Candidiasis is a fungal infection

The most common is *Candida Albicans*.

Immuno Suppression – Major factor in Candidiasis.

Candida organisms normally lives in places such as the mouth, throat, gut, and Genitals.



Acute Candidiasis



1. Pseudo-Membranous/Thrush



2. Erythematous / Erosive/
Antibiotic Stomatitis

Chronic Candidiasis



3. Hyperplastic/Candidal Leukoplakia

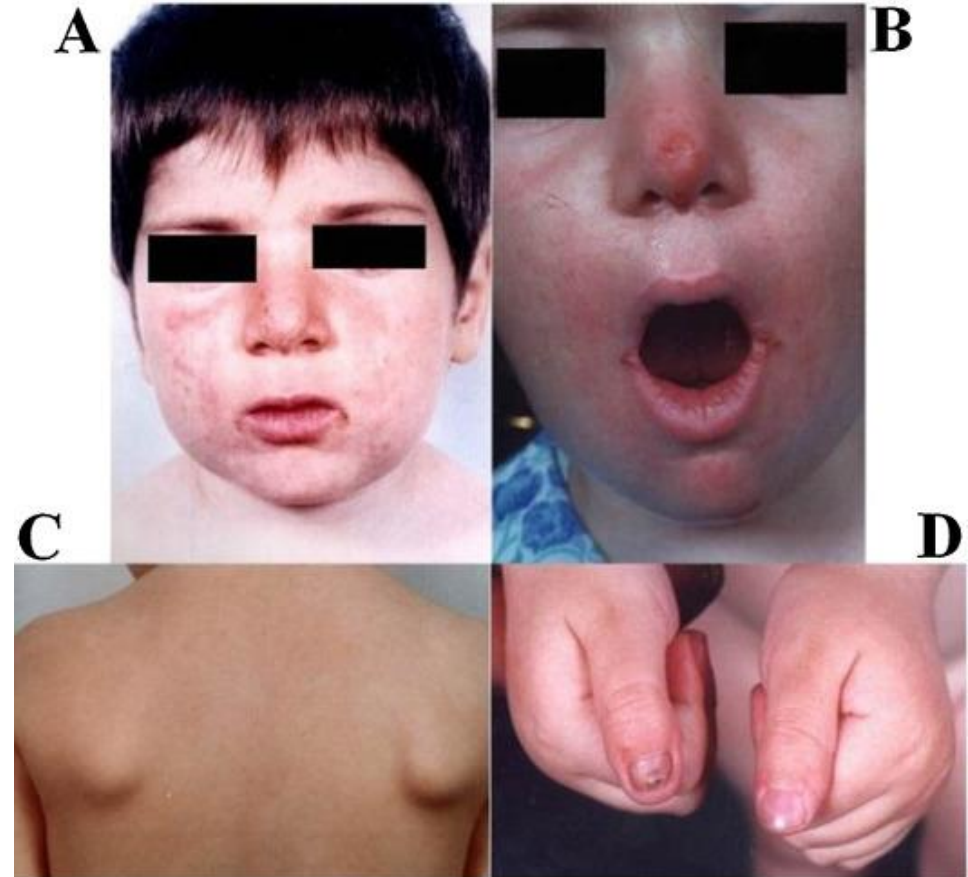


4. Erythematous/Erosive/
Denture Stomatitis

Mucocutaneous



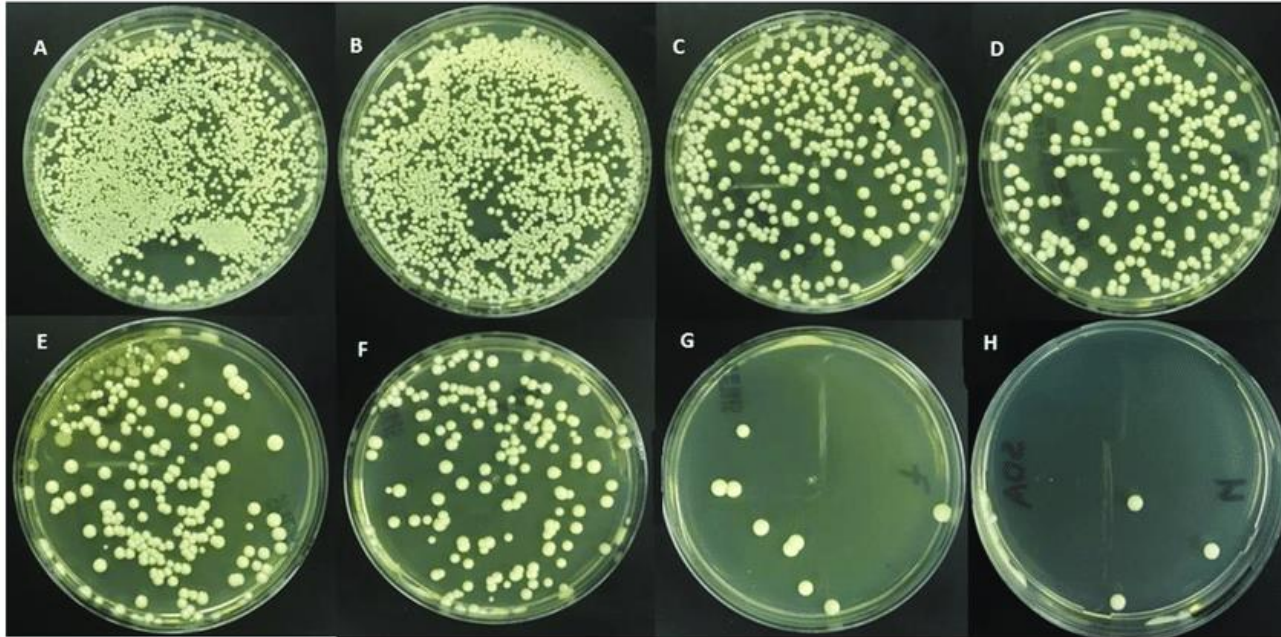
5. Localized



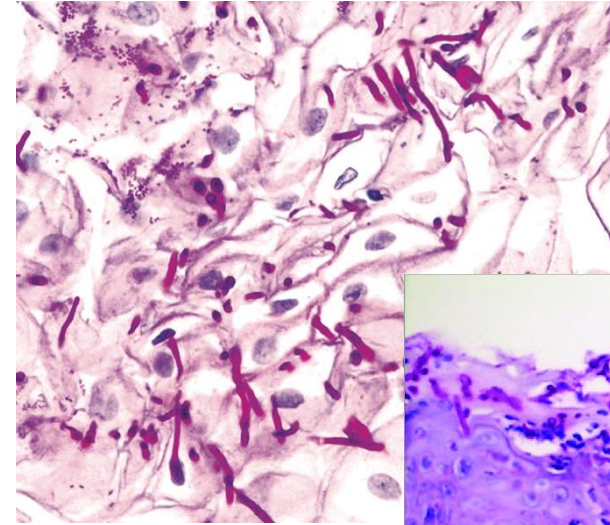
6. Familial

7. Syndrome Associated

Culture

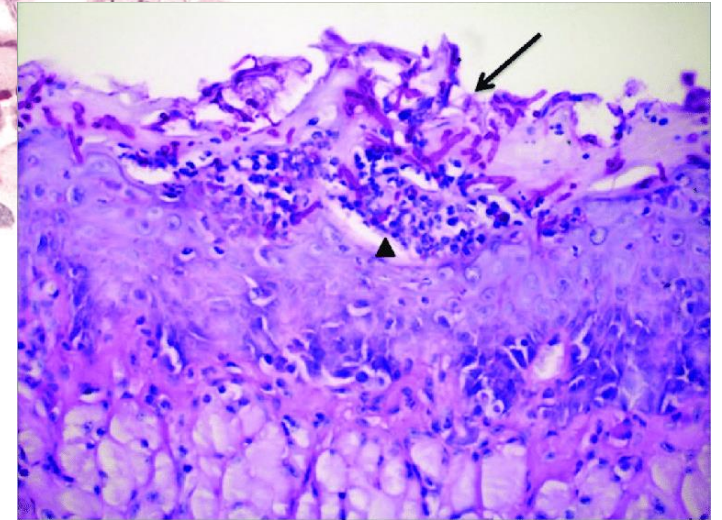


Microscopy



PAS Stain -
Cytology

PAS Stain -
Histopathology



Treatment

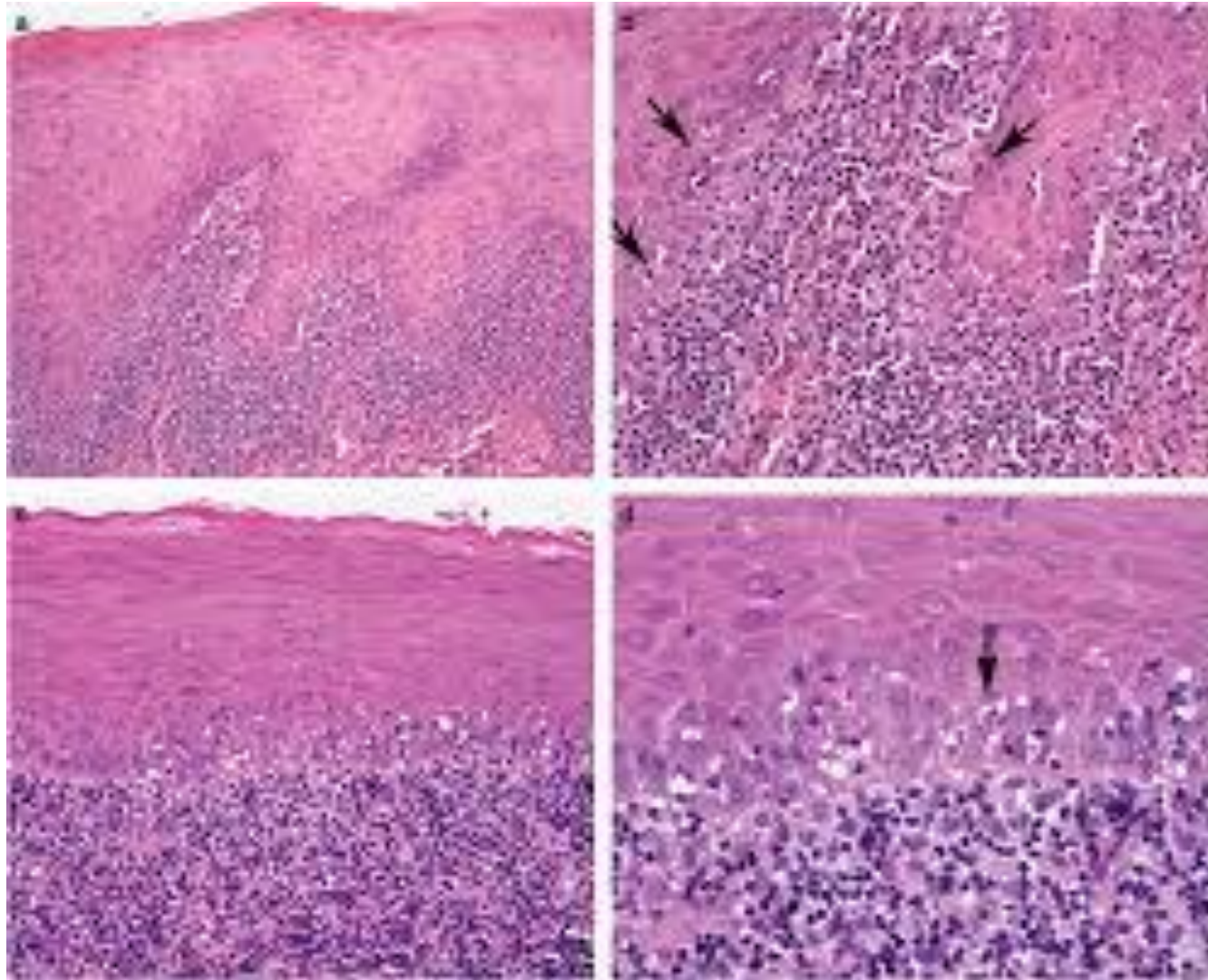
- Improvement in general health condition
 - Anti-fungal Medication

LICHEN PLANUS

- Mucocutaneous Disease
- Unknown cause (Immunological Process)
- 0.2% to 2% of population
- Bilateral white lesions (lace like)
- Association with squamous cell carcinoma



Etiopathogenesis



Initiators: Stress, Dental materials, infectious agents and medications

Initiating factor



Focal Release of cytokines



Retention of T-cells

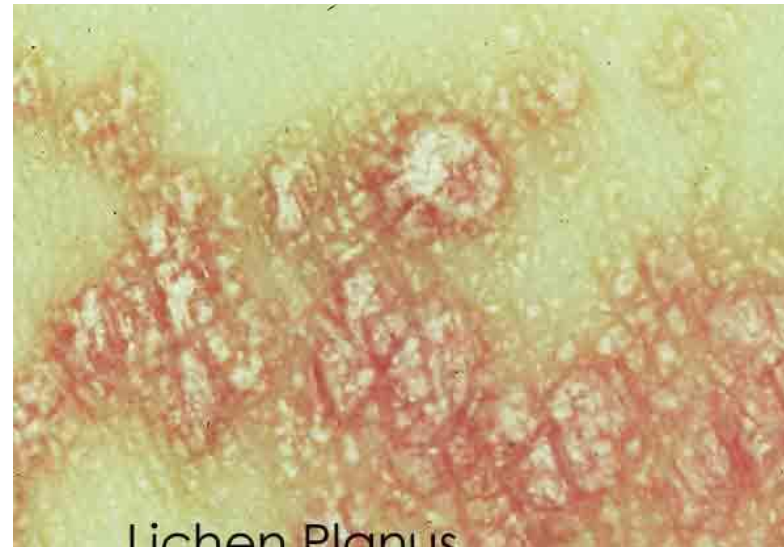


Cytotoxicity of Basal cells



Hyperkeratosis

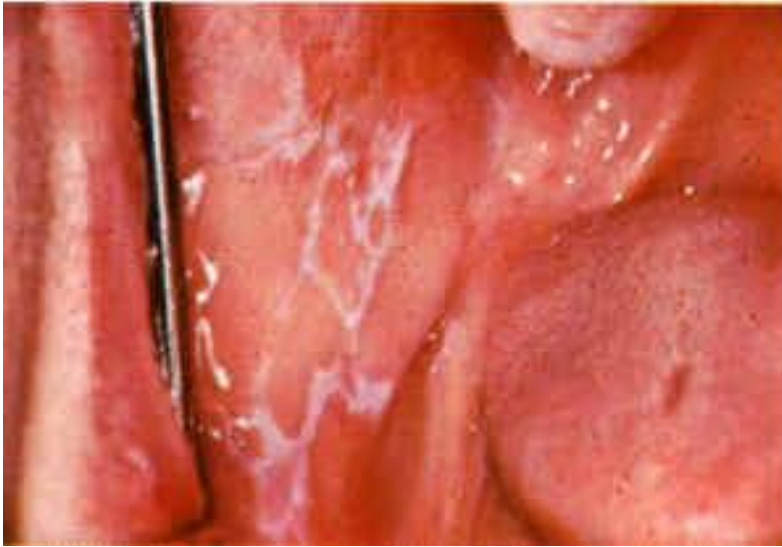
Cutaneous Lichen Planus



- Pruritic
- Planar
- Purple
- Polygonal
- Papules

Oral Lichen Planus

Reticular



Plaque Type



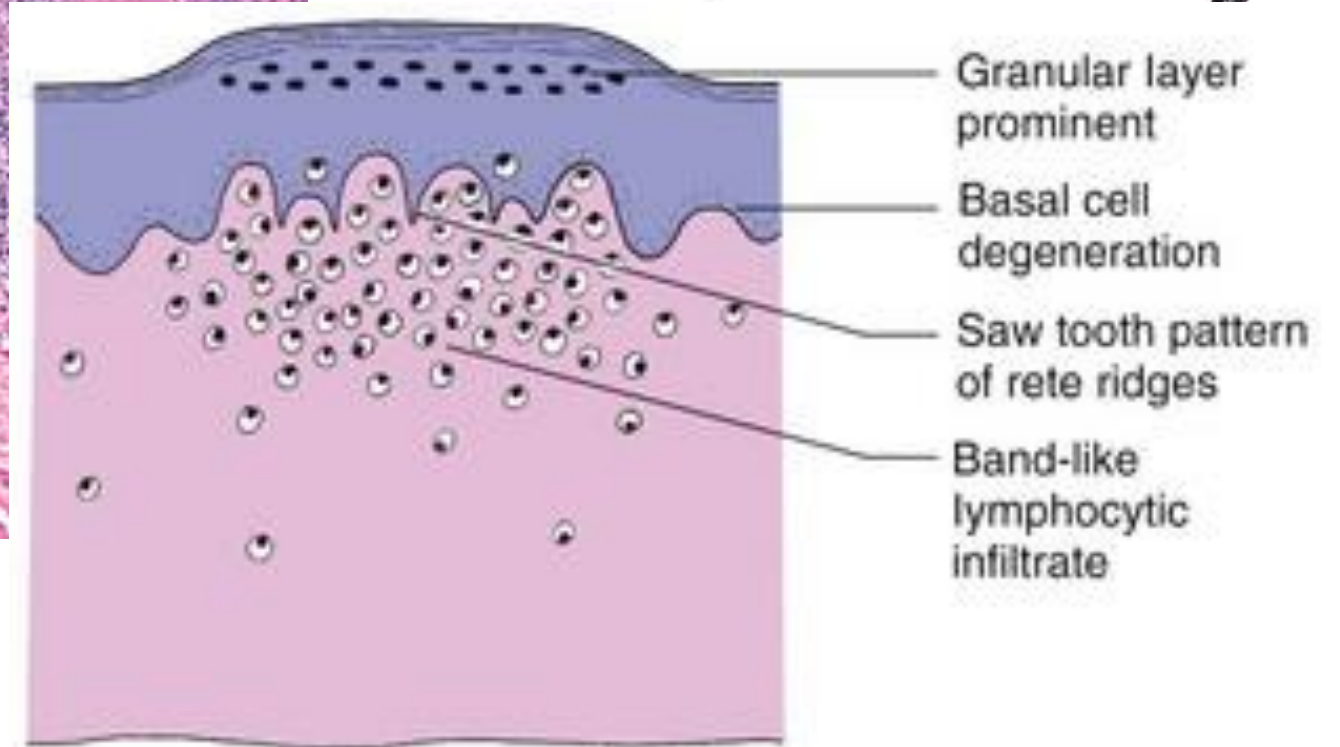
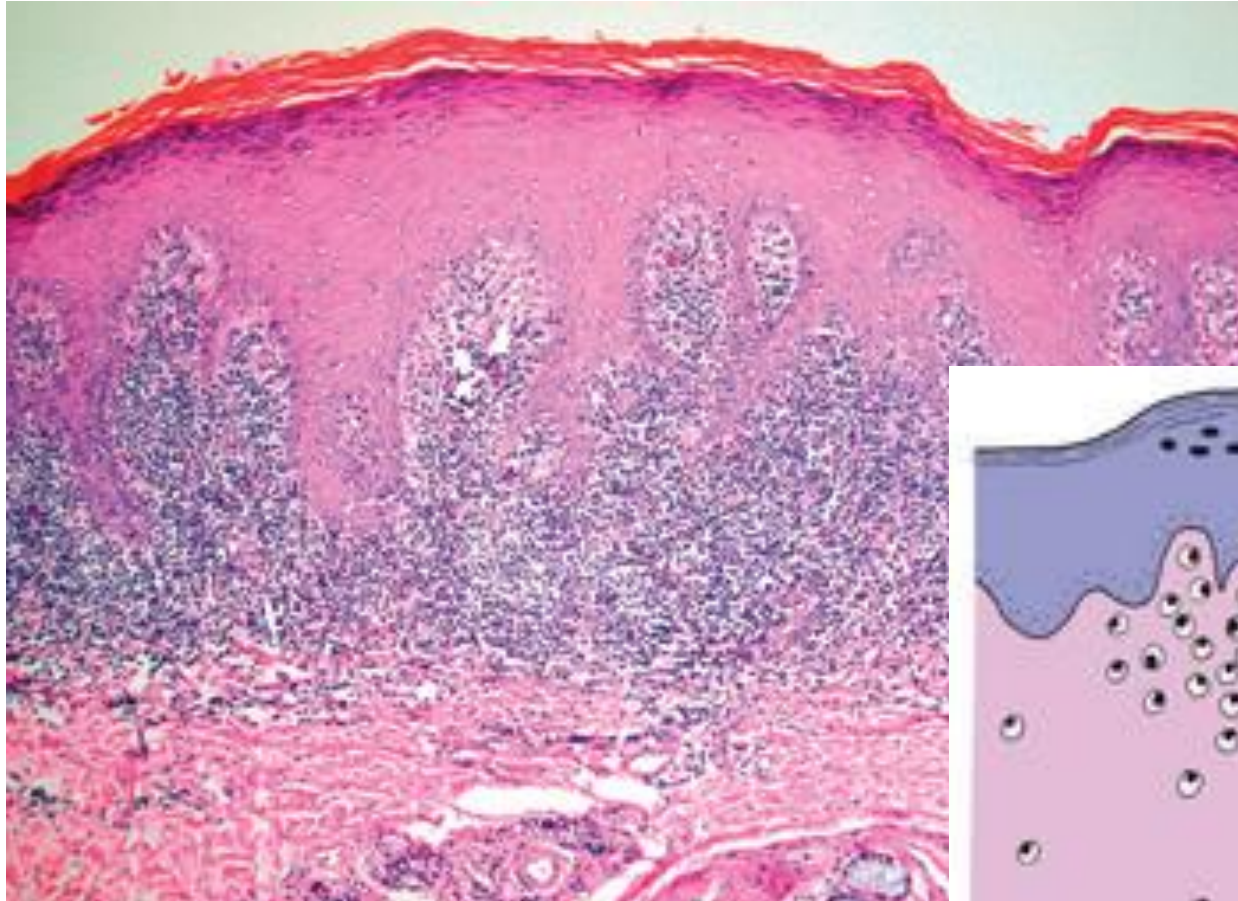
Erythematous/Atrophic



Erosive



Histopathology



Treatment & Prognosis

Currently there is no cure; Varying degree of success with:

- Oral Retinoids
- Oral and Topical Steroids
- Immunosuppressant Medications
- Hydroxychloroquine
- Antifungal Medications

- **Malignant Transformation (???)**:

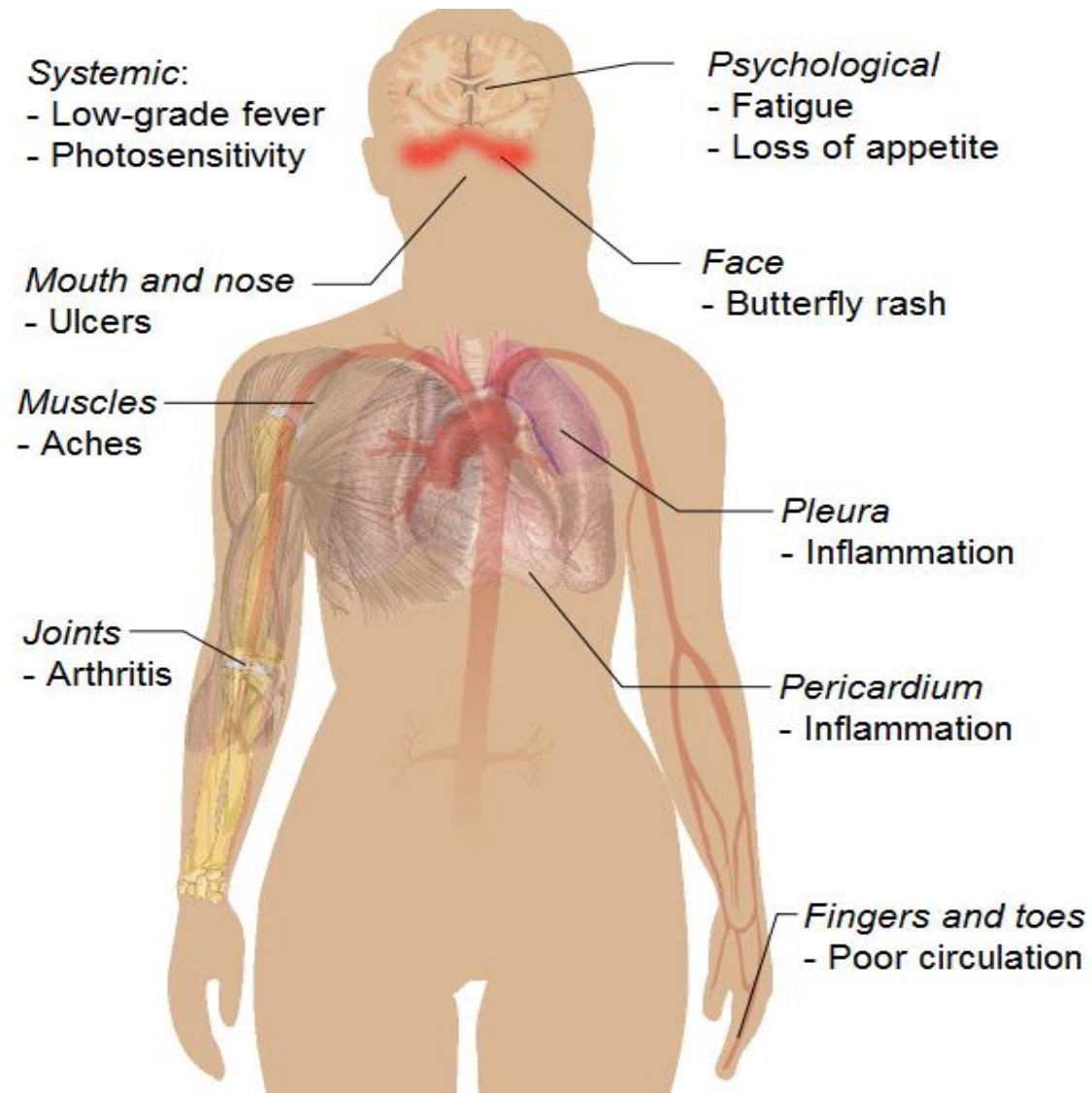
Erosive and Atrophic forms of OLP

- Periodic observation is Mandatory.

LUPUS ERYTHEMATOSUS

- It is a category / a collection of diseases with similar underlying problems with immunity.
- Four main types of lupus exist
 1. **Systemic lupus erythematosus,**
 2. **Discoid lupus erythematosus,**
 3. Drug-induced lupus erythematosus and
 4. Neonatal lupus erythematosus

Systemic Lupus Erythematosus (SLE)





Discoid Lupus Erythematosus

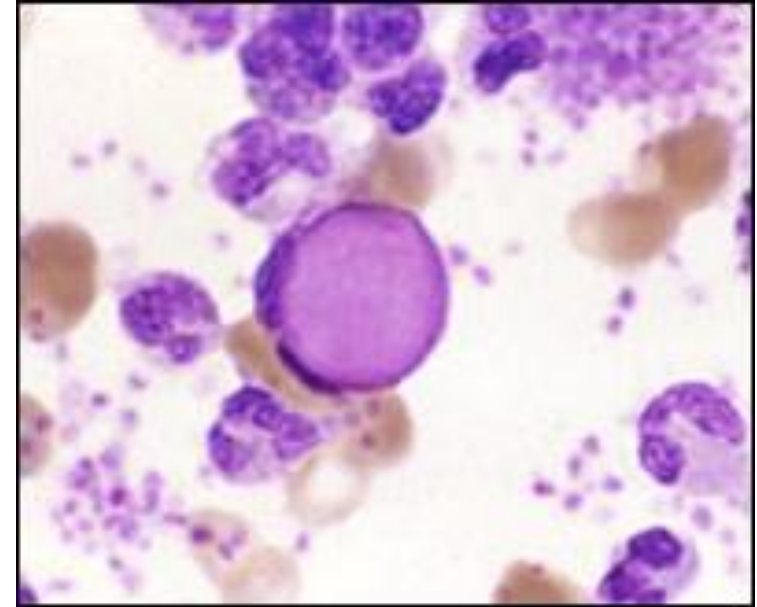


Diagnosis

Serology: antinuclear antibodies

Histopathology: similar to lichen planus

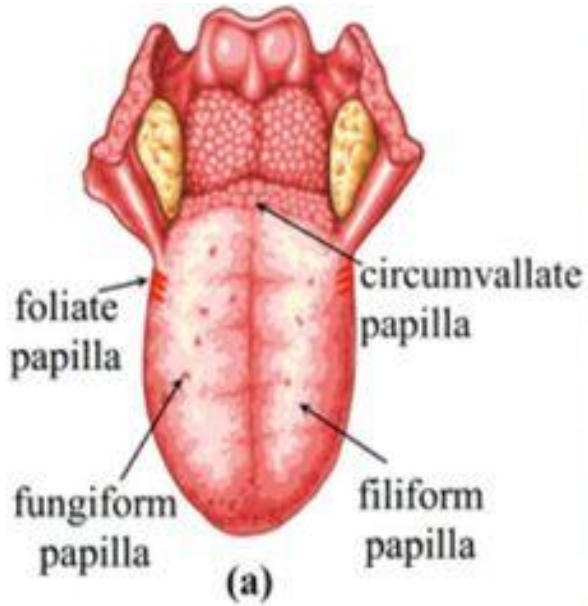
Cytology: shows LE cells in peripheral smear



Treatment

- Immunosuppressive Agents
 - Corticosteroids
- Antimalarials / Sulfones

GEOGRAPHIC TONGUE



- Exact cause of Geographic Tongue is unknown, but it may be linked to genetics, stress, hormonal changes, or vitamin deficiencies; it is seen among people with hypersensitivity, psoriasis, or fissured tongue.
- It causes smooth, red patches on the tongue with white or light-colored borders, giving the tongue a “map-like” appearance (Migratory Glossitis).
- No treatment is needed because the condition is harmless and often improves on its own.

Assessment

A magnifying glass with a black handle and a silver-colored rim is positioned over the word 'Assessment'. The lens of the magnifying glass is centered over the letters 'm', 'e', and 'n', making them appear larger and more prominent than the rest of the word. The word 'Assessment' is written in a large, orange, serif font. The background is a plain, light gray.

- **REFERENCE TEXT BOOKS**

1. Regezi & Sciubba: Oral Pathology - Clinicopathologic Correlation;
2017 Elsevier (7th edn).
2. Shafer's oral pathology – 2015 Elsevier (6th edition)